

Warm-Up Homework #0

CS 4120-5120
Spring 2025

Due: Wed, Jan 22, 2025
30 points

A NOTE TO ALWAYS REMEMBER: IN THIS COURSE, in order to receive full credit, you are expected to show your work leading to the answer, not just the answer alone (even if correct). An incorrect answer by itself would get zero points, but the same answer, accompanied by a reasonable justification showing the right approach might get partial credit. A correct answer alone by itself will not get full credit since you are expected to show your understanding of the approach by showing your work.

1. [4 pts.] Suppose your programming language does not support multi-dimensional arrays, and so you have decided to simulate a 2-dimensional array. Let us assume all array indexes normally start at 1, not 0. For example, suppose you have an array A of 2 rows and 4 columns, which you think of as

A[1,1] A[1,2] A[1,3] A[1,4]
A[2,1] A[2,2] A[2,3] A[2,4]

This two-dimensional (2D) array is actually in a linear array B, so the elements of the first row are stored first, and the elements of the second row are stored next, and so on. [This is called “row-major order”.] In other words,

A[1,1] A[1,2] A[1,3] A[1,4] A[2,1] A[2,2] A[2,3] A[2,4] are stored in:
B[1] B[2] B[3] B[4] B[5] B[6] B[7] B[8] respectively.

Thus, you simulate the 2D array by mapping the pair of indexes into the single index of the linear array B, which actually stores the data. For example, A[1,2] is stored in B[2] and A[2,3] is stored at B[7]. Assuming that A has R rows and C columns, devise a formula for k such that B[k] stores A[i,j], given the row index i and column index j. [**Hint:** Think about it - your formula for k would obviously depend on i and j, but could *possibly* depend on R and C, as well! It is ALWAYS important to get a feel for the “parameters” of a formula, i.e., variables that comprise the formula.]

If you KNOW the rows and columns you could set it to wrap at the width or height value

2. [3 pts.] Do the same as in #1, i.e., assuming that A has R rows and C columns, devise a formula for k such that B[k] stores A[i,j], given the row index i and column index j, but this time assuming that the elements of A are stored in B in “column major” order. As an example, for the 2x4 matrix A in the above example, the order will be as follows:

Ex. Here k =

A[1,1] A[2,1] A[1,2] A[2,2] A[1,3] A[2,3] A[1,4] A[2,4] are stored in:
B[1] B[2] B[3] B[4] B[5] B[6] B[7] B[8] respectively.

3. [4 pts.] Suppose we have a distance chart as in a map. Such a chart shows the distance for all pairs of cities (assuming the distance is the same in either direction between a given pair of cities). If we have a total of n cities, we just need to show (n-1) distances from the first city (city #1), but only (n-2) distances from the second city (city #2) to all but the first city (i.e., city #3 to city #n)! (That is because the distance from the first city to the second has already been recorded

for the first city, and we have already assumed that the distance is the same in either direction.) Then, there will be one fewer distance to record for each of the remaining cities, with no distance left to be recorded for the last city! Also, in this specially-shaped distance array A, due the symmetry of distances discussed above, the distance $A[i,j]$ from city i to city j and the distance $A[j,i]$ from city j to city i will be the same, and will actually be stored corresponding to $A[i,j]$ if $i < j$, or corresponding to $A[j,i]$ if $j < i$.

a. Suppose the distance chart A for n cities were to be stored in a single-dimensional array B. Clearly, $A[1,2]$ through $A[1,n]$ will be stored in $B[1], B[2], \dots$ all the way through $B[n-1]$, then $A[2,3]$ through $A[2,n]$ will be stored next in $B[n], B[n+1], \dots$ all the way through $B[2n-3]$, and so on. Assuming there are n cities, devise a formula for k such that the distance $A[i,j]$ is stored in $B[k]$.

b. Certain values or combinations of values are invalid for i and j . Obviously, any combination with either i or j greater than n is invalid! Explain why a combination where i and j are equal is also not valid.

c. For a total of n cities, give a formula for how many distances need to be stored.

4. [3 pts.] Suppose you perform the following operations on a stack. What will be the output?

Push 1 1 → 21 → 321 → 41321
 Push 2
 Push 3
 Push 4
 Pop and output the popped item
 Pop and output the popped item
 Push 5 521 → 6521
 Push 6
 Pop and output the popped item
 Pop and output the popped item
 Pop and output the popped item

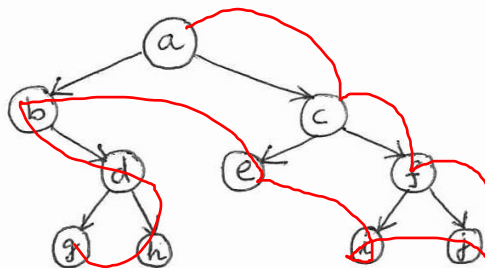
Out
 4
 3
 6
 5
 2

At this point is there an overflow or underflow on the stack? Are there any items left over on the stack?

"overflow" yes, 1

5. [3 pts.] Perform a post-order traversal on the following binary tree, i.e., show the order in which the nodes are visited during a post-order traversal:

ghdabeifca



6. [4 pts.] Keeping in mind that the following traversal strings both correspond to the same tree, construct the single binary tree that gives rise to both the following pre-order and in-order

traversal strings. Note that each letter corresponds to a node. You may approach this problem in a "brute-force" or trial-and-error manner, but an algorithmic approach might be more rewarding!

Pre-order traversal: a b d c e g h f i

In-order traversal: d b a g e h c i f



7. [4 pts.] For $n \geq 0$, prove the following, using mathematical induction:

$$\sum_{i=1}^n i = n(n+1)/2$$

8. Can you devise an algorithm for the task mentioned in problem #6? (A yes/no answer is acceptable this time, although normally it would be considered a "smart-aleck" response and therefore unacceptable. A good answer might possibly make up for little slips in the rest of the homework.)

I don't know an exact algorithm, but you could create an array to compare the two on what values would be where ex, in order starts w/ root

$$\frac{n(n+1)}{2} + \frac{2(n+1)}{2} = \frac{(n+1)(n+2)}{2}$$

$$n(n+1) + 2n + 2 = (n+1)(n+2)$$

$$n^2 + n + 2n + 2 = n^2 + 2n + 1 + 2$$

$$n^2 + 3n + 2 = n^2 + 3n + 2 \quad \checkmark$$